Day 13 2/11/26

283- Move Zeroes

Input: [0,1,0,3,12], Output: [1,3,12,0,0]

Approach: Two Pointer

Use one pointer index to indicate the position where the next non-zero element should be placed.

1. Start index = 0
2. Traverse the array using i
3. Whenever nums[i] != 0:
   - put nums[i] at nums[index].
   - Increment index
4. After all non-zero elements are placed, fill the remaining positions with 0.

Time Complexity: O(n)
Space Complexity: O(1)

Code

```
class Solution
public void moveZeroes(int[] nums) {
    int index = 0;
    for(int i=0; i<nums.length; i++){
        if(nums[i] != 0) {
            nums[index] = nums[i];
            index++;
        }
    }
    while(index < nums.length){
        nums[index] = 0;
        index++; }
}
```

Description | Accepted | Editorial | Solutions | Submissions

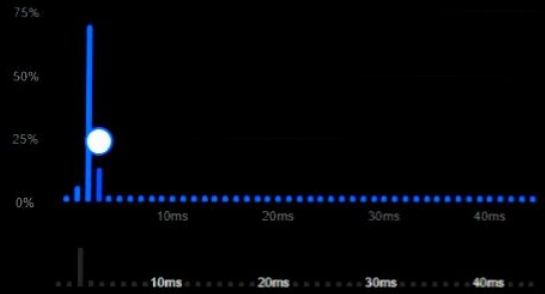
All Submissions

Runtime

3 ms | Beats **23.09%**

Memory

47.33 MB | Beats 79.38%



Code | Java

Code

Java | Auto

```java
class Solution {
    public void moveZeroes(int[] nums) {
        int i =0;
        for(int j=0; j<nums.length; j++){
            if(nums[j]!=0){
                int temp = nums[i];
                nums[i] = nums[j];
                nums[j] = temp;
                i++;
            }
        }
```

Saved | Ln 6, Col 17

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

nums =

`[0,1,0,3,12]`